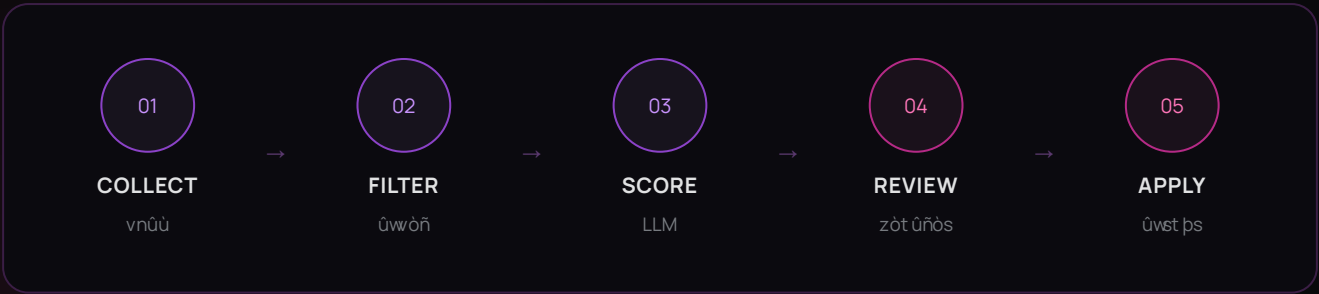


LOCAL-FIRST AUTOMATION / PRODUCTION ARCHITECTURE

á pvnòu ún6 öûsxu òúwrfp6

Öüÿpwòswùmj LLM-ûffòúsmjsúúwùút pùxòu° òúwst psp,
úmt 5 òmòu úvwtpnòpûümv ún6öúvwñsmñ production.



- hh / HeadHunter
- T / á-Òmús
- ĩ gá úòòsv
- VK / VK Team

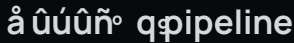
CODE SNAPSHOT main @ 3d31520	UPDATED 14yòúw6nù62026
CONTROL explicit approval	OBSERVABILITY read only

á ûöòùœúþò

1. á þvvòum mñsúðúúu mmpnffò	11. HH Agent Observatory
2. 2 vûsûxúúñûòñm6mûÿpwòsvxùm	12. Windows Scheduler þshared AgentLock
3. X ûöòt ttö múú° ÿþœppúóúú° qffpst	13. Runtime state þt ûdp
4. Discovery: úvþvwúúpsmðú SQLite	14. CI/CD þproduction deployment
5. Evaluation pipeline	15. Safety invariants
6. æðp5 uðþvûúúñûðpwòt ttú° ò	16. Security þppñðnvú° òúòumþzòúþ6
7. Permission model þu mûfflûxvþpnffþ6úvst þsúñ	17. Operational runbook
8. HH apply: instant apply þvûúúñûðpwòt ttúðúúvt ò úvst þsm	18. úmûvwmúòúþpwúþ6
9. Telegram — control plane	19. ævþpuòúpt úvttúúvt òvwmúúú PDF úvð04.09.2026
10. Resume Raise	20. þ úsxuòúw* ú6öúu

HH Agent — t ùsmt tñ° q Windows-first modúwáûpvs mñmñw . ý úa ûnpñmòwñkñmñvpppñsòvñsùt tñpñvñwñzùpsññ, ùwñmñ° ãñmòwñzòñpòbùññòbùññöyöò6fi pò, ðñmòwñzùññù° u ðhard-filter reject ùmññùññmñLLM-mñòt t 6ffp5 , ðñ° ùñt ù6òw vñwñxwñzùpñùññmñùx5 ãLLM-ùñfòùsx, ðññöñpñmòwñzòp5 u ò, ðññwññpñwñvacancy-aware vññùññññöñpòwòt tñùñòðñpvtñù, ùñsmñp° ãñmòwñzòsñùòðññffpñññTelegram pñwññmññt 6òwññt tñññzòññwñst psp, ðññwññù° òñññùñññòññùpñt ðñt tñpññmñwòt tt áñùñ, ðñfòùsñmñzòñffñbùpòðñùt tñpññmñwòt 6pñsite-specific apply adapters ùmññòòòò° ðñññmñpñ° u ðñññwññù, ðwññ° pñuòùòùpòðññwòy òqñvñmñmñwñzòññùt ðññòpñmñòwñññùùòùòwññmññt mñfòùsñpñt pñmñpñòffñbùpòðññmñwññmñsx.

Ùúúvù	âðsxfi òòûñîðòóúpò
Discovery	HH.ru, Yandex Jobs, VK Team, â-Ôús
Filtering	hard filters + LLM appeal öt 6avüüú° ýreject + role/domain policy
Scoring	t úsnt ttú° qOllama, structured output, evidence guard, management policy
Resume	ñ° núúûüý þt 6/ùòp5 u ò, single-resume experiment þwòt òuòwþ6
Cover letter	òòúòèþùxòw6ûíðõúúúúòwúx5 ãñsmúnþ5 þþvüüt tpxòwút tsúûíöwñòúòðòúú° òý nsw
Approval	snoð° qávwst þsavùònxòw6ñúúúúúüt tpûñmvòt tvsvúú° approved
Apply	üwòèt ttú° òaworkers öt 6H, Yandex þK; â-Ôús— discovery only
Telegram	ùòsúuòúömfþp, health/status, ùxzúúqpnúxvs, recovery þJURL → cover-letter flow
Resume Raise	smart supervisor, súwú° qávrú° ñmòw-H wút tsúúúçþòmòuúuxñúòuòþãòvút mwúúúúúð° òum
Observatory	t úsnt túñ6read-only ümòt tLIVE / ANALYTICS / RESUME
Targeted Hunt	üòþvsvázspñýüömþúòt òñnúwúúúúúúwsvññsúuüüþp
CI/CD	branch → PR → CI → main → CD → Octopus → production PC



Apply ϕ to $\tilde{w}$ to get $\tilde{u}$ pipeline:

g wũ ãmpòòt òúpòàùùpúffüüpnt tũû. Pipeline u ûcèwã ùnpùmwtípáuffòúpñnwteĩns mívppnòpòmüxvsmúwst psũñ. Apply supervisor zpwnòw wût tsû6ñúû ãmpùòffbüúx5 ãzùòòött Observatory úpzòòúàòà òù6òwpsüüöst 5 zõũsãmú° u wût tsûámzwòúbò.

Vacancy

āmúpwaíúúunt ppûñmúx5 3msnúp5 pꞤpöòúwpzúúvwtꞤpvnwúúpsmꞤmúúúpzòvsn6pꞤpöòúwpzúúvwtꞤ- [source +](#)
 external_id; hh_id 3wÿÿmúóúót 64egacy-vûñuòvwpuúvvpꞤH.

Evaluation

Úwùþ6áffòúúsxüýùnóðw6öt 6axöpwmxöúòqúnŷüö6w6fwúúñ qscore þdecision, role/seniority/domain/
responsibility fit, gaps þred flags, ùòûuòúöñffþ6,xòòùòþùñíñúúúòþxüüüññöþwòt ttúò,ñ nùnúú qresume key þ
uòwòñúú òmatching.

Application

ãumúpwüüt tpuñmwòt tvsuòuòffbuòpòpoperational lifecycle ûvst psm

á wmwv	á u° vt
notified	smúwúzsrmúúsnpmúmhTelegram, ùòfffbúp6òfi òáòw
approved	Üüt ttpúñmwòt tēñúúánpùòffpt áwst ps
applying	worker únzmt ánnúmwxs
waiting_captcha	úxcòúáxzúúqCAPTCHA / vntwccòwzòt úñòsm
applied	xvüöýáúüöwñòúccòú, pmüüt úóú: applied_at
manual_required	nñwúmvpsmòpúümvúúávwñúñpt mvtptccòwzòt úñòsm
apply_error	wòýúpzòvsn6úffpnsrmòúáúüöwñòúccòúúúqávwúññsp
skipped	Üüt ttpúñmwòt tēüúüxvvtpt áñmsnúvp5
company_blacklist	súu ünúp6ávwúññt òúmñblacklist flow

Y vuuññûqsafety-ûùpúffpû:ý nsw approved x/pt tíúòò-ùùòò° öxfi ööü.LM decision þáúþúnznöwöññùò
 ûùt tpûñmòt tvsùò-ùmpùòffþúpò. Evaluation üùvt òòwüüüüvñwöw6ñùúwòsvñwü þþvñwùþòq,ñùòöññùù° u ÿsù° w u
 veto-gate.

4. Discovery: ùwòvñwúpsmòû SQLite

4.1 HH.ru

Production collector pñuxvsnòw6zòùòp: `hh_collect_optimized.py`. Ý úáwvzpwñmpûñ q: `hh_collect.py`, áwú vñyùnmòwòòû browser/session logic p watchdog.

âòsxfi n6wvñwòdp6:

- persistent Playwright profile `browser-profile`;
- ùùññòùsm HH-vòvvp6áùòòòöwñùùu;
- ùòùvùunt ttú° òáòsùuòúömfppáñnùmmw ññ5 w6áùòñ° up;
- fallback target-role search pñuxvsnòw6wùt tsùòvt páòsùuòúömfppáùòòvñwúúú;
- target-role queries ùùpùùpvpppùx5 w6;
- úmfallback SERP úzòñpöúúáòffòt òñ° òáitle úwòpññ5 w6áùáwvñ° vp6smùwúsp;
- xòöwvñyùnmòúú° òññsmùvpáñwñ5 w6ñ SQLite òmòáùùpáùvt òòx5 fi òu pññpvñùpp browser process;
- child collector sùwùùt pùxòw6 watchdog, sùwùù° qvñùvñnòúáñwñùñpwt pññpvffòòòòòññùùffòvññ.

Òmpûñ° òpùnzòùp6:

```
HH_RECOMMENDATION_PAGES=3
HH_FALLBACK_MIN_NEW_FROM_RECOMMENDATIONS=10
HH_DELAY_BETWEEN_VACANCIES=7
HH_COLLECT_NAVIGATION_TIMEOUT_MS=30000
HH_COLLECT_WATCHDOG_SECONDS=120
```

`hh_collect_optimized.py` áùùùt úpwòt ttùùxuòúttfñòwvñwñ° òádefault delays uòöxwvñmùpffmu pfpñùùvñmupáù 8pá0wòsúúöwñùwñòùùù,òvt páùpáùòáùòùùùòòòòù° environment variables.

Qvt pñHH-vòvvp6áùùwxyt m pipeline úòáùùòuòúòwáùòvñùunt ttú° qvñùùáñ° zú° u áùùpvsùu: HH collection ùùùùxvsnòw6,áùùt tpùññwòt ttáùùt xzmòwáùùòòxùùòöòùùpò,ññùùùùùmvpññ° òpñwúúpspáòrt ttùòqffñ6áñnùmmùwsm ùùùòùt çñ5 w6.

4.2 Yandex Jobs

`sources/yandex.py` ññ° pññòw6zòùòp: `collect_careers.py`. Ávñwúúpsppùt pùññmùáwòùxòpý career collectors. Apply ññ° ùùt úòw `yandex_apply_worker.py` wùt tsùáòt 6 `approved` p wùt tsùáùp: `YANDEX_APPLY_LIVE=true`.

4.3 VK Team

`sources/vk.py` wùnpùñòwññmùvpvpñ/VK Team. Worker xuòòwvpñùùt úòvttcontact/about/social fields, pñùxòwtt ññ° nùmùùòáùòp5 uò,áñnùmmw ññwttconsent, ùòmùpzòúùùöwttùxúx5 CAPTCHA páñvùùpùññwttvùòffùx5 ùwùññsx,áffpñsx pñùstructural form change.

4.4 â-Òmús

`sources/tbank.py` — discovery-only source. Ý úáwúòwñòwstatic pagination p dynamic Playwright discovery, xuòòw ùñyùòpwt lazy-loaded catalog, òòòxùt pffpùxòwññmùvpáùùùUID, ùwñññ° ññwñinactive/closed páùùpuòúòw

management/product/project title policy. Transient HTTP errors will be handled with a bounded backoff.

â-Ômúsámuòòòúúûúòúûmúfflúxwpppúxòw6ñautomatic apply: úwòòt túúôûs `tbank_apply_worker.py` 9òqzmvúòw

4.5 Fault isolation career-pvwûzúpsûñ

`collect_careers.py` smüxvsnòwYandex, VK pã-Ômúsuòpmñpvpuû.Ÿ ffpnsmdöúôûfvpwüzúpsmuòsöt œím
Úvwúmñt pñmwttúvwnt ttú ò; pipeline üüüöüt cæbwámmñvxvçwoupsömúó up,súwùò ò xönt úvtúút xzpwtt

5. Evaluation pipeline

âòsxfi pqqüù6öûsûnùmnûwsp:

```

Hard filters
↓
LLM appeal для appealable reject
↓
Structured evaluation (Ollama)
↓
Evidence guard
↓
Deterministic management policy
↓
Resume matcher / selector
↓
Persistence + Telegram eligibility

```

5.1 Hard filters

Hard filters úòæú° æt6æffñóúíwòzòúp6æizòñpöüúáúøfföt òñ° ýñĩrmúvnpq.æútþwpsmúmvrffþúóúmwnrs,æwún° CTO/CIO/IT Director/Head/úxsúññíðpwèt þæðòmùmwmu òúwññ,úúmùmmñt òúpqfau òæú° òsenior technology leadership titles úóñ° ünñnt þawít ttsûðp-omúòüüþñ° zúúgý ûùuxxt þùññsb.

5.2 LLM-muòt t 6ffp6hard reject

Yòsmæ qreject úsúúzmòt óú. ð t 6appealable cases **HardFilterAppealReviewer** úúúúvpwLm úúúúòòt pwt,úò
6ñt 6òw6ð ðhard reject false negative.

- verdict: `confirm_reject` or `override_reject`;
- confidence: `0.0..1.0`;
- default threshold or `override`: `0.75`;
- unprompted structured response;
- prompt `tuúwòòttúúwòònxòwèxpwr ñmwòòvnmüònmúú òàuxvvpò/múotpqyspò` titles, C-level, Director/Head þ
ùòòttúú òâûn6pmúúûvvp,úòâûüþòxu ññ6smüöþömxûú w

Úvwûùpzòvnbòreject uôoxwüòuòûffòubñmwtv6ûwòttú° u ð backfill flow.

5.3 Structured scoring

Òmpùññ6ý ùùuxt m

```
role_match          * 0.35 +
seniority_match      * 0.20 +
domain_match         * 0.15 +
responsibility_match * 0.30
```

Evaluation pòðwzòùòpž ùsnt ttú° qđOllama pstructured schema. Evidence guard ùùùñòù6òwzwnñ° ãùòđùùpùòòw6 ùmñmñvñp5 ðđùòwñòùòđòùú° òý mñw ðùùý pt 6. Management policy pñwòu ðùùpuòù6òwzòùòu pùpùññmú° ò smùttòùú° òđùùññpt mñùñòùýđLLM ùòpxt ttwmm

5.4 LLM resilience

Qvt pđGPU pñù6wpt pđOllama ãùòuòùùúđòòùvñxùòù, scoring/appeal úòđùòññmfi mñ w6ñđ ùòđ° qđreject. Žmñvñp6 ùvññw6ðending ðauùòðwñ° wttñnùmñmñmñt òòx5 fi pu ðùùýùòùu. ð t pùú° qđpòùùññ° qđ [process_vacancies.py](#) ùòđùòmùpzòùmñfi pu ðwall-clock timeout; timeouts ùvñññ w6ñmñùññòùòùsùòwñ° ýññòfflùpýđùòùñffpq.

6. ðòp5 u òðvñùùùññòpòt ttú° ò

Resume selection ùùùpvýùòpòwñsite adapter. Ž° nùmú° qđ ùsnt ttú° qđPDF ãñt pòpùòw6zòùòp: [app/application_assets.py](#) ðđmñvñmñpññòw6ñmñsubmission source of truth.

á ùùùññòpòt ttùòđùpvttu ù:

- vùpòññw6ñùòñùsùòwñx5 ãmñvñp5 ;
- vñ6p° ãññwñùt ttsùòò òñmñwñ° òđùòwñòùòđòùú° òý mñw ðùùý pt 6 ðvñòññmùp6u ðùùt p;
- úòðùòùòðwñplaceholders;
- úòññ° òxu° ãññwñù° wñwòspt pñùvñpòùp6;
- vùýùmñ6òw6ññEvaluation/Application context ðpñwòu ðvùùt ttpxòw6ñworker'ùu.

Vacancy URL → cover letter zòùòpTelegram

Z ùt ttùññwòt ttuùòðwñùpvt mñttv° t sxmññmñvñp5 . Flow smùùpñpùòwñURL, ùùùòòò 6òwñHH/Yandex/VK, ùù ãùpuùòđùvñpñvùùt ttpxòwñòðvñùmñùùùòEvaluation letter, pùnzòđùùt xzmñwñvacancy content ðòùòùpùòw ùññòđùpvttu ùzòùòpž ùsnt ttùòđùòp5 uò, preferences pđOllama. P t 6ñòppññòwñ° ýñpublic HTTP(S) vñqñññgeneric fetch path ùòùmùpzpññòlocalhost/private/service networks, ùòvññòmñwñ° òđùùñw ðđòsùùwñùt pùòù° ò redirects.

7. Permission model ပာမီရှင်များကို ပြန်လည်စစ်ဆေးခြင်း

User approval — authoritative

`Application.status == approved` မှတ်တမ်းများကို စစ်ဆေးခြင်းကို ပြုလုပ်ရာတွင် `business approval gate` ကို ဖြတ်ကျော်ရန် လိုအပ်သည်။

ပုံစံများကို ထိန်းချုပ်ခြင်း

```
YANDEX_APPLY_LIVE=true
VK_APPLY_LIVE=true
```

switches — ကွက်တိုက်မှုများကို ထိန်းချုပ်ရန် `false` သို့မဟုတ် `true` ဖြစ်နိုင်သည်။

Source routing

- HH worker ကို `approved` ဖြစ်စေရန် HH rows ကို `source` ဖြစ်စေရန်;
- Yandex worker — `approved` Yandex;
- VK worker — `approved` VK;
- အလိုအလျောက် apply route.

No blind retry after submit

တစ်ခေါ်တစ်ခေါ် `success` ဖြစ်ပါက `submit` ကို ပြုလုပ်ရန် လိုအပ်သည်။

8. HH apply: instant apply ပုံစံများကို ထိန်းချုပ်ခြင်း

HH ပုံစံများကို ထိန်းချုပ်ရန် `instant apply` ပုံစံများကို ထိန်းချုပ်ရန် လိုအပ်သည်။

အောက်ဖော်ပြပါများကို

- `instant apply` ပုံစံများကို ထိန်းချုပ်ရန်;
- `source-specific selectors` ပုံစံများကို ထိန်းချုပ်ရန်;
- `worker` ပုံစံများကို ထိန်းချုပ်ရန်;
- `success-banner` ပုံစံများကို ထိန်းချုပ်ရန်;

- üüðwñòùöóùpòáúpvttumípfí ðw6áúúáúvòò ttú° u «success markers pt þáúúáúú6ñþñffñbuvx6áúwñnùmæøù úux wðsvwxáúpvttumñúóðinput/textarea;
- üüþáúúúðúúþúnzúúú áúòpxt ttwðvñvút ttpxðw6«guarded verification/retry logic, úúáúúñwñúúúúðöqvwñþò pñúúðfí ðúúúwñu, æðòáúúáúúúöwáúpömwttöxnt tt
- övt þáúwt psáúúzúúáúwñunñt óú, múpvttuúáúúðwñòúðöpwttúúöxómt úvtt, status vwmúñþpw6: manual_required »v» applied_at :á/vpnvòmzðvñwñúáúûñfi nòw«úwt psðvñttáúpvttuúáúwñòxðwñúúúñòúsp».

DOM/regression tests, delayed post-apply UI, submit targeting guarded retry.

9. Telegram – control plane

Production entry point: `telegram_bot_entry.py`.

Ùúuúôm	Z ùñòòóùpò
/health	system/Ollama/runtime/queue health
/status	ý ùúúñ° òàùvù6úþ6þbreakdown approved queue
/run	pnûxvsñòwpipeline vòqzmv
/new	unresolved/manual-required snùwûzsp+ùúñ° òàòòóúòúöñffpp
/stats	vvnþvvpvsmApplication statuses

/new **úònt ûspùxòwnûwm**

[illegible]

Recovery pmanual_required

/new ၁/únznt mñûpñunfi nõw manual_required, ၁mòu unresolved notified, ၁úvt òzòòúòñmñt 6òw òòqvwñpòt tuúúúñ òòòòúòòòffpp. Best-effort notification ú manual_required ၁úòòòòpñmñp5, súuúúp5, ၁úpzpúx, Application ID ၁súúüsxúwsùò wò6ñmñp5.

Telegram Watchdog

Yw0ot tUm6scheduled task HH Agent - Telegram Watchdog uñUñòù6òwaTelegram ùmpñu púxvw.Ý úñvwt òøpñmòw runtime heartbeat, PID þý mwpzòvspqprocess command line. Default stale threshold — 180 vòsxüö.Zùþpnñpvffbu heartbeat pt þpvzòpúxñffbu process watchdog üòòpñuxsmòwscheduled task, üòòðñmüpòt túúxnþññ6stale telegram bot entry.py process.

- UI úòpñxvsmòwworkers þáðáúòú6òwbusiness state.

Ý vñú° wttt úsnt ttúú: `http://127.0.0.1:8765`.

12. Windows Scheduler þshared AgentLock

Scheduled task	ænnvúwn/trigger	Entry point	Ý vñòúúúvþ
HH Agent - Pipeline	smø° òðznm	<code>background_pipeline.py</code>	HH session guard, runtime heartbeat
HH Agent - Apply	smø° òð0 upúxw	<code>background_apply.py</code>	zþwòwútt tsúapproved queue
HH Agent - Resume Raise	smø° òð5 upúxw	<code>background_resume_raise.py</code>	heavy browser only near <code>next_due_at</code>
HH Agent - Telegram	üüþðogon	<code>telegram_bot_entry.py</code>	pythonw, restart policy, IgnoreNew
HH Agent - Telegram Watchdog	smøx5 upúxw	<code>telegram_watchdog.py</code>	stale heartbeat / PID recovery
HH Agent - Dashboard	üüþðogon	<code>dashboard</code>	local read-only UI

Pipeline, Apply þResume Raise þvüút tpx5 wñfi þqAgentLock. g vñúòúüþñútt 6òwñúúý t þswx5 fi þu ðbrowser jobs úmúwñttúúmt t òt ttúþáúðffwttöúxóðúxóæzòðþúüý þt þ, UI þt þnetwork state.

Production deployment þvüút tpxòwñúwöðAgentLock, zvñ° Git update úòáúüþvñüðpt áúvòúòðþúððñvþñúq browser operation.

13. Runtime state þt úð

Runtime JSON:

```
data/runtime/pipeline.json
data/runtime/apply.json
data/runtime/resume_raise.json
data/runtime/telegram.json
data/runtime/telegram_watchdog.json
```

Ý vúññú° òðt úð:

```

logs/pipeline_supervisor.log
logs/collector.log
logs/careers_collector.log
logs/processor.log
logs/apply_dispatcher.log
logs/apply_supervisor.log
logs/apply_worker.log
logs/apply_worker_attention.log
logs/yandex_apply_worker.log
logs/yandex_apply_worker_attention.log
logs/vk_apply_worker.log
logs/vk_apply_worker_attention.log
logs/telegram.log
logs/telegram_watchdog.log
logs/resume_raise_supervisor.log
logs/resume_raise_worker.log

```

Browser profiles, `.env`, SQLite DB, runtime JSON logs – it's not that operational data but it's still useful.

14. CI/CD for production deployment

It's a bit more complex than the PR-first flow:

```

feature branch
  ↓
pull request
  ↓
GitHub Actions CI
  ↓
merge to main
  ↓
CI on main
  ↓
GitHub Actions CD
  ↓
Octopus Deploy
  ↓
HH-Agent-PC
  ↓
C:\hh-agent (fast-forward only)

```

CI

Windows / Python 3.12 pipeline is quite simple:

- `git diff --check`;
- Python compileall;
- syntax validation `deploy/octopus_deploy.ps1`;
- xvfb + Chromium for DOM regression tests;

- core `unittest` suite.

CD / Octopus

```
Zûvt òxvüòffüûòûCl úm main CD vûpöñröwrelease/deployment ñOctopus Deploy project HH Agent öt 6
environment Production . Target — Windows PC HH-Agent-PC zòùòpPolling Tentacle.
```

```
deploy/octopus_deploy.ps1:
```

- `git pull --rebase` Pipeline/Apply/Resume Raise;
- `git checkout -b feature` AgentLock;
- `git commit -m "feat: Git update + validation window"`;
- `git add .` tracked local changes;
- harmless untracked files `git rm --cached .`;
- `git checkout -b feature` feature branch `main`;
- `git push --set-upstream origin main`;
- `git pull --rebase` Python sanity checks;
- `git push --force` Telegram `git push --force` `origin/main`;
- `git push --force` post-update validation failure `git checkout main` `git push --force` `origin/main`;

Octopus üòüöñw6commit SHA, üü0üuxproduction deployment uüæüxñ6pmwtsüúúüòwú° uGit revision.

`âvùòffll° q&GitHub CD ùpúnzmòwzûp müüvúmd deployment ñ&Octopus vùpö mú.â pünt tt° q&target-side status úxœúúwu ûwùòvttñ&Octopus: target u ûcœw ümwtæxœúüvt ôú súúzmúb6&GitHub job.`

15. Safety invariants

gwpšüüñmpt mwpwr5 w6zmwt5 mñypwòswùò pšöüt öüü müòòöñmwtöüóy mswùùpúó:

1. **Source separation.** Worker ùòòùt ààùpmmpmwtúizòùòòttòùòxóuôsource.
2. **User approval is authoritative.** approved ãipúnzmòwñûíùòòámpùòfffbúpòóùít ttpúñkìmwòt 6.
3. **External live switch.** Yandex/VK final submit wùònXòwãûîwñòwwñx5 fi bq * _APPLY_LIVE=true .
4. **No blind retry after submit.** ÿòòùùùpúnzú° qáùòpxt ttvfwáòóùùpñüòöpwswáùíñwùùùqáùwùùnñisò.
5. **Cover-letter confirmation ≠ application confirmation.** ð t 6 ñHH 0wáùámpú° òóùùsnprwòt ttvñkì
6. **Local resume is the submission source of truth.** Site adapter ùòóùùpòxu° ãñwòùòxóuôqresume asset.
7. **Evaluation history remains auditable.** ávwùùp6áùòòönt 6òw6áùùöþæòüñvvñkì
8. **AgentLock ün6pmwòt óùòt 6áùùý t pswx5 fi pyðbrowser jobs þproduction deployment.**
9. /new áùòáùòùòüpv° ãñwòùòfffbúp6. á úñûvvwnññt þññwøunresolved þóùñnñt 6òwáùùñ° òóñwùúzsp.
10. â-Ômúsáùòáautomatic-apply source, áùùsmúòáùpömúdedicated adapter.
11. **Observatory read-only.** Dashboard ùòóùùòèwáùòù6wttbusiness state.
12. main áùòù6òw6áùòùòþbranch + PR. Production üüt xznòwáùùt tsùáùùùfffböfffbpqCl súö.

Շնչարկային օգնություն օգտագործող օգնականներ

```
data/runtime/apply.json
logs/apply_dispatcher.log
logs/apply_supervisor.log
logs/apply_worker_attention.log
```

Եթե հարկ է դրա՝ օգնականներին օգնություն ցուցաբերելու համար, ապա պետք է օգտագործել `manual_required` ֆիլդը Application ID-ի մեջ: Օգնականները կգտնվեն օգնականների ցանկում, որոնց համար 6 ֆուլ ժամանակահատվածում օգնականներին օգնություն ցուցաբերելու համար:

Շնչարկային օգնություն Telegram

```
data/runtime/telegram.json
data/runtime/telegram_watchdog.json
logs/telegram.log
logs/telegram_watchdog.log
```

18. Մոնիթորինգի օգնականներ

<code>.github/workflows/</code>	CI + CD
<code>app/</code>	evaluation, DB, policies, resume, Targeted Hunt
<code>sources/</code>	Yandex / VK / Т-Банк collectors
<code>dashboard/</code>	FastAPI + Observatory UI
<code>deploy/</code>	Octopus deployment logic
<code>doc/</code>	system and feature documentation
<code>tests/</code>	regression and unit tests

```
hh_collect.py
hh_collect_optimized.py
collect_careers.py
process_vacancies.py
```

```
background_pipeline.py
background_apply.py
background_resume_raise.py
```

```
apply_dispatcher.py
apply_worker.py
yandex_apply_worker.py
vk_apply_worker.py
```

```
telegram_bot_entry.py
telegram_new_background_patch.py
telegram_watchdog.py
```

```
resume_raise_worker_v2.py
resume_raise_schedule.py
```

19. ~~æwî~~ ~~φpu~~ ~~òúpt~~ ~~ûnttûûvt~~ ~~òæw~~ ~~nùûôû~~ PDF ~~ûw~~ 04.09.2026

á wnú° qPDF nút tfffbáòáúwúnært production. Øæwòsxfi x5 ãbòùvp5 æûnmñt òú° φφvüümñt òú° :

- Observatory LIVE / ANALYTICS / RESUME þóôûread-only núÿpwòswxúm
- dashboard telemetry öt 6hard filters þæsingle-resume experiment;
- LLM appeal öt 6æüüúú° ÿhard-filter reject þφvñúþzòvþqbackfill;
- ùmvffþòóúúñ6áúûþ wþpsmsenior IT leadership titles;
- `hh_collect_optimized.py`, ámúúþqSERP gate þáúûñ° òáúûmñpt mHH collection;
- heartbeat pipeline ñûñúòò6æót þúúúôûcollect/process þæú6wþòáúñfi òóúæwall-clock cap vþòúúûñúôûprocessing batch;
- ý úúûññ6: `/new`, æwúúñ6ñúû tfffbáòænt úspùxòw `/health` φæùxqþòæTelegram commands;
- Ûwòòt ttú° qæTelegram Watchdog vññwúmwþzòvþu ærestart þññþvffboûñúwñ
- smart Resume Raise schedule: wake-up smæ° òææþúxwñòþáúúvñú6úúúôûφmüxvsmChromium;
- production CI/CD zòùòþæGitHub Actions + Octopus Deploy + AgentLock;
- commit SHA ñæOctopus deployment;
- hardened production deploy: fast-forward only, protection tracked local changes, rollback validation;
- ñwñkt ttú° qæHH instant-apply flow, oðòæüüúûñúöþwòt ttúúòáúþst ñö° ññòw6áwòòt ttúúáúvt òáúòp5 uò;
- üúûñòùþæáúúúòwúôûætextarea/submit öt 6post-apply letter;
- guarded retry/verification vüüúûñúöþwòt ttúúôûφáúvòòt ttú° q `manual_required`, ævt þáúvst þsòvñtt, mþvttu úáò üúöwñòùæòúú;
- regression tests öt 6æobserved HH DOM variants;
- ñwñkt ttún6æmúwñsource routing þæsafety invariants.

20. ~~φ~~ ~~ûsxu~~ ~~òúw~~ ~~æ~~ ~~ö~~ ~~ûu~~

- `README.md` — æúñwþþqáúñþúúáúúòòswñ
- `dashboard/README.md` — Observatory þáúòúþzòúþ6æread-only vt ú6;
- `doc/ci_cd.md` — CI/CD þæproduction deployment;
- `doc/single_resume_experiment.md` — single-resume experiment;
- `doc/targeted_hunt.md` — Targeted Hunt;
- `doc/hh_apply_success_detection.md` — æþwòùþþáúüöwñòùæòúþ6æHH apply;
- `LICENSE` — æþfòúþþ6áúúòòswñ

HH AGENT / rudenko.one

Local-first automation · explicit user approval · observable production · safe external actions.